Annex II. Debt Sustainability Analysis

The updated debt sustainability analysis (DSA) confirms Togo's moderate risk of external debt distress and heightened risk of overall public debt distress —unchanged from the previous DSA published in April 2017. However, there are significant downside risks to growth and fiscal prospects, which if materialized, would likely result in a deterioration of Togo's risk of external debt distress. Togo's external debt remains vulnerable to adverse shocks: on exports stemming from low external demand or decline in commodity prices, and on non-debt creating flows. Also, Togo's overall public debt dynamic continue to present heightened vulnerabilities, with the present value (PV) of debt-to-GDP currently standing at twice the prudential level of 38 percent, and remaining above the indicative benchmark through 2024.

The revised baseline macroeconomic assumptions for the present DSA are: real GDP growth is currently expected to be lower in the medium-term and revert slightly above the April 2017 forecast in 2022; inflation is revised down 1 and 0.5 percent in 2017 and 2018, respectively; the primary fiscal balance remains anchored on a surplus of 2 percent by the end of the program and until 2025, when debt declines below the threshold of heightened risk of public debt distress; current account deficit is about 1 percent of GDP lower in 2017, due to a higher than expected decline in imports; and FDI is down 0.4 percent of GDP each year, relative to the previous DSA vintage. Once the costs of the bank recapitalization, estimated at 2-4 percent of GDP, are realized and included in the fiscal accounts, the debt profile will worsen.

Under the baseline scenario, all Togo's external debt indicators continue to remain below their indicative policy-relevant thresholds (Table 1, Figure 1). The PV of public and publicly guaranteed (PPG) external debt is projected at 16.5 percent of GDP in 2016 and will stabilize around this level until 2037. The improvement in the external debt dynamic relative to April 2017 stems from improvements in the primary current account deficit that more than offset the deteriorated FDI. Both ratios of the PV of external debt relative to revenues and to exports continue to remain relatively stable and below their respective indicative thresholds through the end of the projection period. Similarly, debt service measures remain well below thresholds and on a broadly downward trend. Improvements in debt-management practices envisaged in the authorities' ECF-supported program will give further resilience to shocks affecting debt service needs (Figure 1). However, under the historical scenario and several other shock scenarios, several debt indicators breach their respective thresholds in outer years.

Togo's overall public debt dynamics also highlight heightened vulnerabilities, with the debt-to-GDP ratio remaining above the indicative benchmark for a significant part of the projection period (Table 2). Overall public debt is now forecasted at 77.3 percent of GDP in 2017, falling below the 70 percent WAEMU convergence criterion by 2019. By 2037, repayment of arrears coupled with significant fiscal consolidation is expected to significantly reduce domestic debt and total public and publicly guaranteed debt. The analysis highlights the need for sustained fiscal consolidation, improved debt management, and macroeconomic policies to reduce the level of public debt to prudent levels over the medium term.

Table 1a. Togo: External Debt Sustainability Framework, Baseline Scenario, 2014–2037 1/

(In percent of GDP, unless otherwise indicated)

	Actual			Historical 6/ Average	Standard 6/ Deviation	Projections									
	2014	2015	2016			2017	2018	2019	2020	2021	2022	2017-2022 Average	2027	2037	2023-2037 Average
External debt (nominal) 1/	**18.6**	**23.2**	**22.2**			**23.5**	**26.1**	**28.9**	**29.5**	**29.8**	**29.7**		**28.6**	**28.0**	
of which: public and publicly guaranteed (PPG)	17.3	21.2	20.4			21.0	23.4	26.1	26.6	26.8	26.6		25.8	26.3	
Change in external debt	3.3	4.6	-1.0			1.4	2.6	2.8	0.6	0.3	-0.1		-0.1	0.0	
Identified net debt-creating flows	15.8	15.4	9.8			9.0	8.5	7.0	5.7	4.3	3.8		2.6	3.1	
Non-interest current account deficit	**9.8**	**10.9**	**9.4**	**8.3**	**2.5**	**7.8**	**7.7**	**6.8**	**5.9**	**4.9**	**4.5**		**4.6**	**5.2**	4.3
Deficit in balance of goods and services	18.0	22.2	20.0			18.4	18.0	17.1	16.2	15.1	14.7		14.8	15.2	
Exports	39.7	36.3	34.9			33.9	32.9	33.7	34.1	35.0	34.5		32.4	27.5	
Imports	57.7	58.5	54.9			52.3	50.9	50.8	50.3	50.2	49.2		47.2	42.7	
Net current transfers (negative = inflow)	-7.0	-7.7	-7.1	-8.2	1.8	-6.7	-6.7	-6.7	-6.7	-6.7	-6.7		-6.7	-6.8	-6.7
of which: official	-1.9	-1.8	-1.7			-1.7	-1.7	-1.7	-1.7	-1.7	-1.7		-1.7	-1.7	
Other current account flows (negative = net inflow)	-1.2	-3.7	-3.6			-3.8	-3.5	-3.6	-3.6	-3.6	-3.6		-3.6	-3.3	
Net FDI (negative = inflow)	**6.7**	**2.2**	**1.7**	**2.3**	**5.6**	**1.7**	**1.4**	**1.0**	**0.7**	**0.4**	**0.4**		**-0.9**	**-0.9**	-0.6
Endogenous debt dynamics 2/	**-0.6**	**2.3**	**-1.2**			**-0.6**	**-0.6**	**-0.8**	**-0.9**	**-1.0**	**-1.1**		**-1.2**	**-1.2**	
Contribution from nominal interest rate	0.2	0.3	0.4			0.4	0.5	0.5	0.5	0.5	0.5		0.4	0.3	
Contribution from real GDP growth	-0.9	-1.1	-1.1			-1.0	-1.1	-1.3	-1.4	-1.5	-1.5		-1.6	-1.5	
Contribution from price and exchange rate changes	0.0	3.1	-0.5			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**-12.5**	**-10.8**	**-10.9**			**-7.6**	**-5.9**	**-4.2**	**-5.1**	**-4.0**	**-3.9**		**-2.7**	**-3.2**	
of which: exceptional financing	-0.3	-0.4	-1.2			-0.7	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	18.3			20.3	21.1	22.0	21.9	21.7	21.3		19.3	18.3	
In percent of exports	...	...	52.5			59.8	64.1	65.3	64.3	61.9	61.6		59.7	66.7	
PV of PPG external debt	**...**	**...**	**16.5**			**17.8**	**18.4**	**19.2**	**19.0**	**18.7**	**18.2**		**16.6**	**16.7**	
In percent of exports	**...**	**...**	**47.3**			**52.4**	**56.0**	**57.0**	**55.6**	**53.3**	**52.8**		**51.3**	**60.6**	
In percent of government revenues	**...**	**...**	**87.6**			**88.8**	**85.1**	**90.2**	**88.8**	**87.6**	**85.3**		**77.8**	**78.1**	
Debt service-to-exports ratio (in percent)	**2.9**	**3.1**	**5.0**			**4.3**	**3.9**	**3.9**	**4.1**	**4.1**	**4.0**		**3.1**	**3.0**	
PPG debt service-to-exports ratio (in percent)	**2.9**	**3.1**	**5.0**			**4.3**	**3.9**	**3.9**	**4.1**	**4.1**	**4.0**		**3.1**	**3.0**	
PPG debt service-to-revenue ratio (in percent)	**6.4**	**5.7**	**9.3**			**7.4**	**5.9**	**6.1**	**6.5**	**6.8**	**6.5**		**4.7**	**3.8**	
Total gross financing need (Billions of U.S. dollars)	0.8	0.6	0.6			0.6	0.6	0.6	0.6	0.5	0.6		0.7	1.8	
Non-interest current account deficit that stabilizes debt ratio	6.5	6.2	10.4			6.5	5.1	4.0	5.3	4.6	4.7		4.8	5.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.9	5.3	5.0	5.0	2.3	4.8	5.0	5.2	5.3	5.4	5.6	5.2	6.0	6.0	6.0
GDP deflator in US dollar terms (change in percent)	0.0	-14.2	2.2	2.0	10.1	2.8	5.9	2.1	2.2	2.3	3.2	3.1	4.0	4.0	4.0
Effective interest rate (percent) 5/	1.6	1.6	1.7	1.1	0.7	2.0	2.2	2.0	1.9	1.8	1.7	1.9	1.4	1.4	1.4
Growth of exports of G&S (US dollar terms, in percent)	-9.5	-17.5	3.2	7.3	14.2	4.7	7.8	10.1	8.7	10.9	7.3	8.3	9.1	4.2	8.5
Growth of imports of G&S (US dollar terms, in percent)	-7.8	-8.5	0.7	8.2	15.5	2.7	8.1	7.2	6.4	7.6	7.0	6.5	9.5	8.7	9.1
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	8.7	51.0	51.0	50.9	46.1	46.7	42.4	45.4	43.7	44.9
Government revenues (excluding grants, in percent of GDP)	18.3	19.7	18.9			20.0	21.7	21.3	21.4	21.3	21.3		21.3	21.3	21.3
Aid flows (in Billions of US dollars) 7/	0.4	0.3	0.4			0.3	0.5	0.5	0.4	0.4	0.5		0.8	2.0	
of which: Grants	0.1	0.1	0.1			0.3	0.2	0.2	0.3	0.3	0.3		0.5	1.3	
of which: Concessional loans	0.3	0.2	0.2			0.1	0.2	0.3	0.2	0.2	0.2		0.3	0.7	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			5.7	6.3	6.7	5.7	5.5	5.5		5.4	5.4	5.4
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			55.6	73.6	72.9	78.8	76.8	77.6		77.1	76.3	76.8
Memorandum items:															
Nominal GDP (Billions of US dollars)	4.6	4.1	4.4			4.8	5.3	5.7	6.1	6.6	7.2		11.7	30.9	
Nominal dollar GDP growth	5.9	-9.7	7.3			7.8	11.2	7.4	7.6	7.9	9.0	8.5	10.2	10.2	10.2
PV of PPG external debt (in Billions of US dollars)			0.7			0.9	1.0	1.1	1.2	1.2	1.3		1.9	5.1	
(PVt-PVt-1)/GDPt-1 (in percent)						4.2	2.0	2.2	1.1	1.2	1.2	2.0	1.5	1.8	1.6
Gross workers' remittances (Billions of US dollars)	0.2	0.2	0.2			0.2	0.3	0.3	0.3	0.3	0.3		0.6	1.6	
PV of PPG external debt (in percent of GDP + remittances)	...	...	15.7			16.9	17.6	18.3	18.1	17.8	17.4		15.8	15.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	41.2			45.5	48.8	49.4	48.7	46.6	46.3		44.3	51.0	
Debt service of PPG external debt (in percent of exports + remittance	...	...	4.4			3.8	3.4	3.4	3.6	3.6	3.5		2.7	2.5	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Assumes that PV of private sector debt is equivalent to its face value.

5/ Current-year interest payments divided by previous period debt stock.

6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

7/ Defined as grants, concessional loans, and debt relief.

8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 1b. Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017-2037

(In percent)

	Projections							
	2017	2018	2019	2020	2021	2022	**2027**	2037
PV of debt-to GDP ratio								
Baseline	18	18	19	19	19	18	**17**	17
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	18	20	22	24	27	30	**45**	72
A2. New public sector loans on less favorable terms in 2017-2037 2	18	20	22	22	23	23	**24**	27
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	18	19	20	20	20	19	**17**	17
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	18	21	27	26	26	25	**22**	18
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	18	21	25	24	24	23	**21**	21
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	18	22	27	27	26	25	**22**	18
B5. Combination of B1-B4 using one-half standard deviation shocks	18	24	33	32	31	30	**26**	21
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	18	26	27	26	26	25	**23**	23
PV of debt-to-exports ratio								
Baseline	52	56	57	56	53	53	**51**	61
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	52	60	66	71	77	86	**138**	263
A2. New public sector loans on less favorable terms in 2017-2037 2	52	60	65	65	65	66	**73**	97
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	52	56	57	56	53	53	**51**	60
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	52	74	110	106	101	100	**92**	90
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	52	56	57	56	53	53	**51**	60
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	52	68	81	78	74	73	**67**	66
B5. Combination of B1-B4 using one-half standard deviation shocks	52	71	97	94	89	88	**81**	79
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	52	56	57	56	53	53	**51**	60
PV of debt-to-revenue ratio								
Baseline	89	85	90	89	88	85	**78**	78
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	89	92	104	114	127	140	**210**	339
A2. New public sector loans on less favorable terms in 2017-2037 2	89	91	102	104	107	107	**111**	125
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	89	87	95	93	92	89	**81**	82
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	89	97	127	124	121	118	**102**	85
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	89	98	116	114	112	109	**99**	100
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	89	103	128	125	122	118	**102**	85
B5. Combination of B1-B4 using one-half standard deviation shocks	89	111	153	149	146	141	**122**	101
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	89	119	126	124	122	119	**108**	109

Table 1b. Togo: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2017–37 (concluded)

(In percent)

Debt service-to-exports ratio								
Baseline	4	4	4	4	4	4	**3**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	4	4	4	4	5	5	**5**	12
A2. New public sector loans on less favorable terms in 2017-2037 2	4	4	3	4	4	4	**4**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	4	4	4	4	4	4	**3**	3
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	4	4	6	6	6	6	**5**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	4	4	4	4	4	4	**3**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	4	4	4	5	5	4	**4**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	4	4	5	5	6	5	**4**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	4	4	4	4	4	4	**3**	3
Debt service-to-revenue ratio								
Baseline	7	6	6	6	7	6	**5**	4
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2017-2037 1/	7	6	6	7	8	8	**8**	15
A2. New public sector loans on less favorable terms in 2017-2037 2	7	6	5	6	6	6	**6**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2018-2019	7	6	6	7	7	7	**5**	4
B2. Export value growth at historical average minus one standard deviation in 2018-2019 3/	7	6	6	7	8	7	**6**	5
B3. US dollar GDP deflator at historical average minus one standard deviation in 2018-2019	7	7	8	8	9	8	**6**	5
B4. Net non-debt creating flows at historical average minus one standard deviation in 2018-2019 4/	7	6	7	7	8	7	**6**	5
B5. Combination of B1-B4 using one-half standard deviation shocks	7	6	8	9	9	9	**7**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2018 5/	7	8	9	9	9	9	**7**	5
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	42	42	42	42	42	42	**42**	42

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming
an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 2a. Togo: Public Sector Debt Sustainability Framework, Baseline Scenario, 2014–37 1/

(In percent of GDP, unless otherwise indicated)

	Actual					Estimate							Projections						
	2014	2015	2016	Average 5/	Standard Deviation 5/	2017	2018	2019	2020	2021	2022	2023	2024	2025	2026	2017-22 Average	2027	2037	2023-37 Average
Public sector debt 1/	65.1	75.6	81.5			77.3	74.4	70.2	66.7	63.0	58.9	54.7	50.2	45.7	42.9		41.6	38.4	
of which: foreign-currency denominated	17.3	21.2	20.4			21.0	23.4	26.1	26.6	26.8	26.6	26.4	26.2	26.0	25.9		25.8	26.3	
Change in public sector debt	9.4	10.5	6.0			-4.2	-2.9	-4.3	-3.5	-3.7	-4.2	-4.2	-4.4	-4.5	-2.8		-1.3	-0.4	
Identified debt-creating flows	5.3	6.2	5.0			-4.1	-2.4	-5.1	-4.0	-3.7	-4.2	-4.5	-4.8	-4.8	-3.2		-1.6	-0.4	
Primary deficit	5.5	6.8	7.1	4.1	2.3	-0.2	0.0	-3.5	-2.1	-2.0	-2.1	-2.1	-2.1	-2.1	-0.5	-1.7	1.0	2.2	1.1
Revenue and grants	20.7	22.1	21.7			25.3	25.6	25.4	25.5	25.4	25.4	25.4	25.4	25.4	25.4		25.4	25.4	
of which: grants	2.3	2.3	2.9			5.3	3.9	4.1	4.1	4.1	4.1	4.1	4.1	4.1	4.1		4.1	4.1	
Primary (noninterest) expenditure	26.1	28.9	28.8			25.1	25.6	21.9	23.4	23.4	23.3	23.3	23.3	23.3	24.9		26.4	27.6	
Automatic debt dynamics	-0.2	-0.6	-2.1			-4.0	-2.4	-1.7	-2.0	-1.7	-2.1	-2.4	-2.7	-2.7	-2.6		-2.6	-2.6	
Contribution from interest rate/growth differential	-1.7	-2.8	-2.8			-1.9	-2.3	-1.6	-2.0	-1.8	-2.1	-2.4	-2.7	-2.7	-2.6		-2.6	-2.6	
of which: contribution from average real interest rate	1.4	0.5	0.8			1.8	1.4	2.1	1.5	1.6	1.3	0.9	0.4	0.1	-0.1		-0.1	-0.4	
of which: contribution from real GDP growth	-3.1	-3.3	-3.6			-3.7	-3.7	-3.7	-3.5	-3.4	-3.4	-3.2	-3.1	-2.8	-2.6		-2.4	-2.2	
Contribution from real exchange rate depreciation	1.5	2.2	0.6			-2.0	-0.1	-0.1	0.0	0.1	0.0	0.0	0.0	0.0	0.0		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	4.1	4.3	1.0			-0.1	-0.5	0.9	0.6	0.0	0.1	0.3	0.3	0.3	0.4		0.3	0.0	
Other Sustainability Indicators																			
PV of public sector debt	...	...	77.7			74.1	69.4	63.3	59.1	54.9	50.5	46.0	41.4	36.7	33.8		32.4	28.7	
of which: foreign-currency denominated	...	...	16.5			17.8	18.4	19.2	19.0	18.7	18.2	17.7	17.3	17.0	16.8		16.6	16.7	
of which: external	...	...	16.5			17.8	18.4	19.2	19.0	18.7	18.2	17.7	17.3	17.0	16.8		16.6	16.7	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...	...	...	...	...		...	...	
Gross financing need 2/	20.6	27.7	19.6			9.7	9.4	6.8	12.6	13.6	12.1	9.2	9.2	8.8	9.1		9.1	8.7	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	357.3			292.8	271.4	249.2	232.2	216.1	198.5	180.9	162.7	144.4	132.9		127.4	113.0	
PV of public sector debt-to-revenue ratio (in percent)	...	...	411.9			370.0	320.5	297.1	276.7	257.6	236.6	215.7	194.0	172.1	158.4		151.8	134.7	
of which: external 3/	...	...	87.6			88.8	85.1	90.2	88.8	87.6	85.3	83.2	81.3	79.7	78.6		77.8	78.1	
Debt service-to-revenue and grants ratio (in percent) 4/	73.3	94.7	57.8			39.1	37.0	40.3	57.5	61.4	56.0	44.7	44.4	42.9	37.7		31.7	25.5	
Debt service-to-revenue ratio (in percent) 4/	82.6	105.9	66.6			49.4	43.7	48.0	68.6	73.2	66.8	53.3	52.9	51.1	44.9		37.8	30.4	
Primary deficit that stabilizes the debt-to-GDP ratio	-3.9	-3.7	1.1			4.0	2.9	0.8	1.4	1.6	2.0	2.1	2.3	2.4	2.3		2.3	2.6	
Key macroeconomic and fiscal assumptions																			
Real GDP growth (in percent)	5.9	5.3	5.0	5.0	2.3	4.8	5.0	5.2	5.3	5.4	5.6	5.8	6.0	6.0	6.0	5.2	6.0	6.0	6.0
Average nominal interest rate on forex debt (in percent)	1.7	1.7	1.9	1.1	0.7	2.2	2.4	2.3	2.1	2.0	1.9	1.8	1.8	1.7	1.6	2.1	1.6	1.5	1.5
Average real interest rate on domestic debt (in percent)	3.0	1.5	1.8	-0.1	3.6	2.7	2.2	4.0	3.6	4.7	4.6	4.3	3.5	3.0	2.7	3.7	2.7	1.6	2.9
Real exchange rate depreciation (in percent, + indicates depreciation	11.1	13.4	3.2	2.6	7.9	-10.4	...	...	...	...	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	-0.1	2.7	2.4	2.9	3.4	0.9	1.5	1.6	2.0	2.7	3.2	3.6	3.8	4.0	4.0	2.0	4.0	4.0	4.0
Growth of real primary spending (deflated by GDP deflator, in percer	8.0	16.3	4.9	3.0	5.4	-8.6	6.8	-9.8	12.3	5.5	5.1	5.9	6.0	6.0	13.3	1.9	12.4	6.0	7.2
Grant element of new external borrowing (in percent)	...	...	...	...	...	8.7	51.0	51.0	50.9	46.1	46.7	46.1	45.9	45.7	45.6	42.4	45.4	43.7	...

Sources: Country authorities; and staff estimates and projections.

1/ [Indicate coverage of public sector, e.g., general government or nonfinancial public sector. Also whether net or gross debt is used.]

2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.

3/ Revenues excluding grants.

4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.

5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

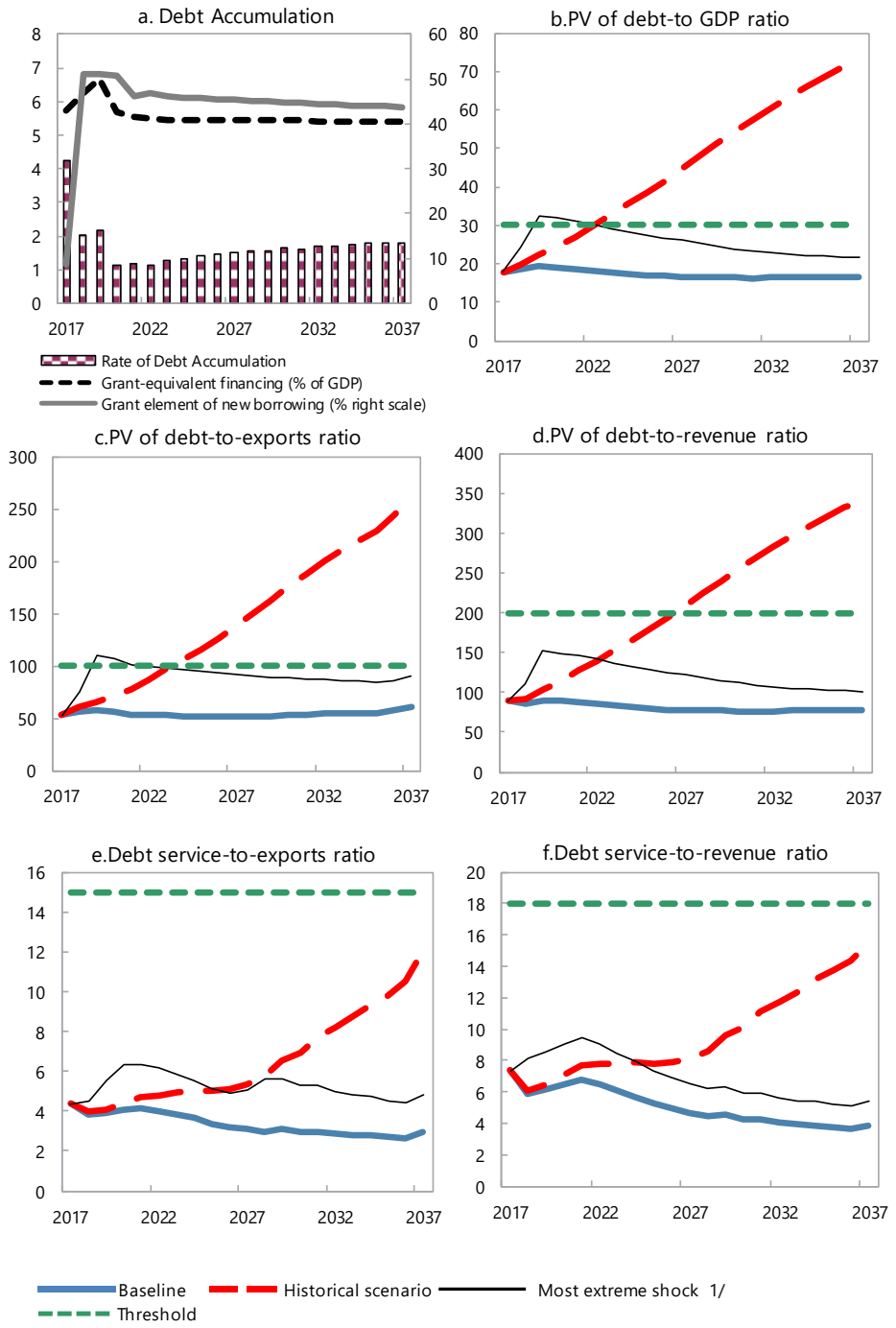
Table 2b. Togo: Togo: Sensitivity Analysis for Key Indicators of Public Debt 2017–37

	Projections							
	2017	2018	2019	2020	2021	2022	2027	2037
PV of Debt-to-GDP Ratio								
Baseline	74	69	63	59	55	50	32	29
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	74	72	72	72	72	71	65	60
A2. Primary balance is unchanged from 2017	74	69	66	63	60	56	39	20
A3. Permanently lower GDP growth 1/	74	70	64	60	57	53	37	42
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-20	74	71	67	64	60	56	39	38
B2. Primary balance is at historical average minus one standard deviations in 2018-201	74	74	75	70	65	60	39	32
B3. Combination of B1-B2 using one half standard deviation shocks	74	74	75	70	66	61	41	36
B4. One-time 30 percent real depreciation in 2018	74	76	68	63	59	53	33	27
B5. 10 percent of GDP increase in other debt-creating flows in 2018	74	77	70	66	61	56	36	31
PV of Debt-to-Revenue Ratio 2/								
Baseline	293	271	249	232	216	198	127	113
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	293	283	282	282	281	279	254	228
A2. Primary balance is unchanged from 2017	293	271	258	246	234	221	154	78
A3. Permanently lower GDP growth 1/	293	273	252	237	222	206	146	164
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-20	293	278	263	248	234	218	153	148
B2. Primary balance is at historical average minus one standard deviations in 2018-201	293	290	295	275	257	236	154	126
B3. Combination of B1-B2 using one half standard deviation shocks	293	289	294	276	259	239	162	142
B4. One-time 30 percent real depreciation in 2018	293	297	270	249	231	210	129	106
B5. 10 percent of GDP increase in other debt-creating flows in 2018	293	299	276	258	240	221	143	121
Debt Service-to-Revenue Ratio 2/								
Baseline	39	37	40	58	61	56	32	25
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	39	37	41	60	67	64	46	38
A2. Primary balance is unchanged from 2017	39	37	40	58	63	58	35	22
A3. Permanently lower GDP growth 1/	39	37	41	58	63	57	34	31
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2018-20	39	38	42	60	65	59	35	29
B2. Primary balance is at historical average minus one standard deviations in 2018-201	39	37	41	61	68	62	34	27
B3. Combination of B1-B2 using one half standard deviation shocks	39	37	42	62	68	63	35	29
B4. One-time 30 percent real depreciation in 2018	39	38	42	60	64	59	34	28
B5. 10 percent of GDP increase in other debt-creating flows in 2018	39	37	41	61	65	60	33	26

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Figure 1. Togo: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2017–37 1/

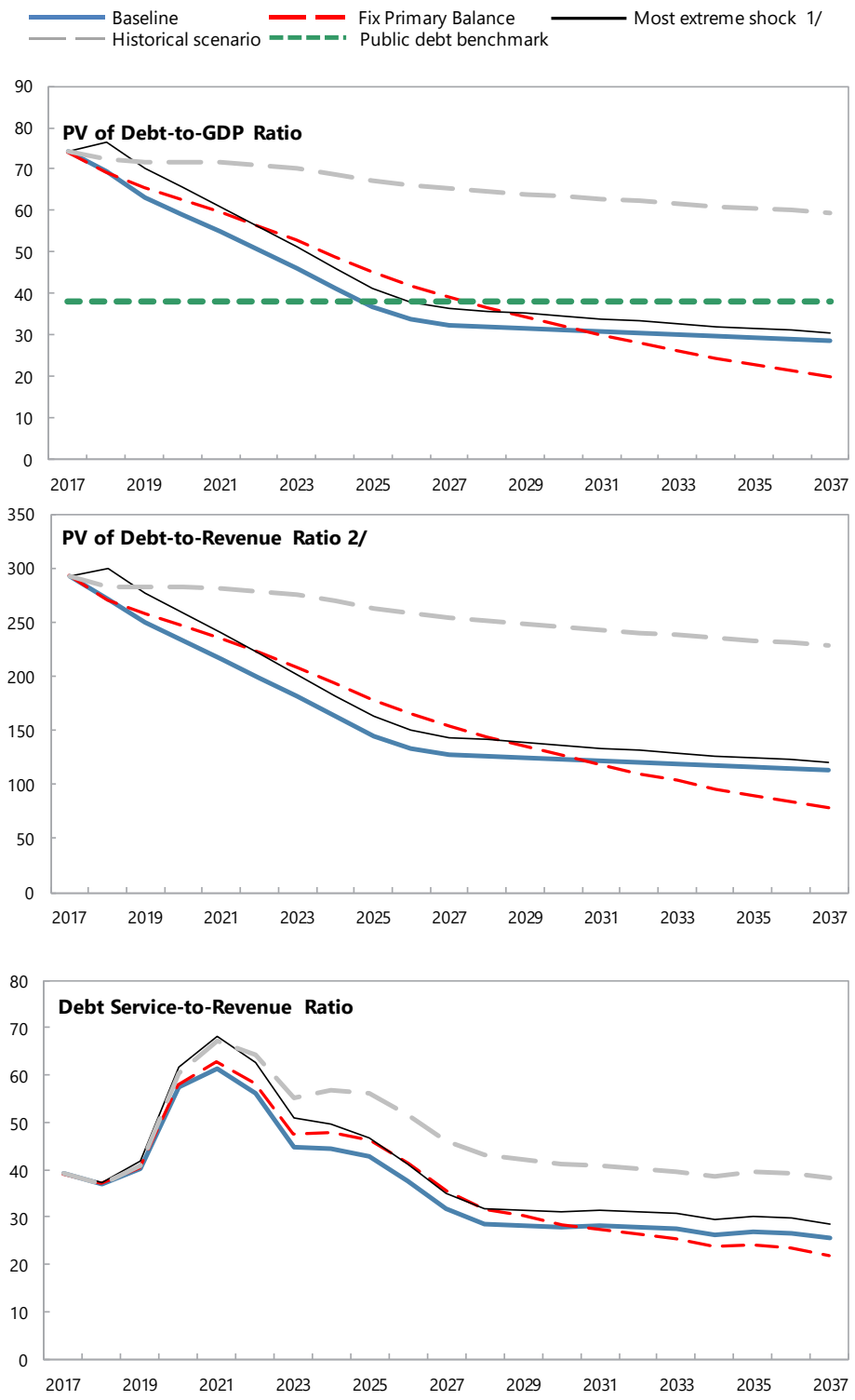
(In percent of GDP, unless otherwise indicated)



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio on or before 2027. In figure b. it corresponds to a Combination shock; in c. to a Exports shock; in d. to a Combination shock; in e. to a Exports shock and in figure f. to a One-time depreciation shock

Figure 2. Togo: Indicators of Public Debt under Alternatives Scenarios, 2017–37 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio on or before 2027.
2/ Revenues are defined inclusive of grants.